

Theoretical Answers

- 1. What is a Support Vector Machine (SVM)?**
A Support Vector Machine is a supervised learning algorithm used mainly for classification, and sometimes regression. It finds the best boundary, called a hyperplane, that separates different classes with the maximum margin.
- 2. What is the difference between Hard Margin and Soft Margin SVM?**
Hard Margin SVM assumes the data is perfectly separable and does not allow misclassification. Soft Margin SVM allows some errors using slack variables, which makes it more useful for real-world data.
- 3. What is the mathematical intuition behind SVM?**
SVM tries to find the hyperplane that maximizes the distance between the classes. The larger the margin, the better the model is expected to generalize on unseen data.
- 4. What is the role of Lagrange Multipliers in SVM?**
Lagrange Multipliers are used to solve the optimization problem in SVM efficiently. They help convert the constrained problem into a form that can be solved using optimization techniques.
- 5. What are Support Vectors in SVM?**
Support vectors are the data points closest to the decision boundary. They are the most important points because they determine the position of the hyperplane.
- 6. What is a Support Vector Classifier (SVC)?**
SVC is the classification version of SVM. It is used to classify data into categories by finding an optimal separating boundary.
- 7. What is a Support Vector Regressor (SVR)?**
SVR is the regression version of SVM. Instead of classifying data, it predicts continuous values while trying to keep prediction errors within an acceptable margin.
- 8. What is the Kernel Trick in SVM?**
The Kernel Trick allows SVM to work on non-linear data by transforming it into a higher-dimensional space where it becomes easier to separate.
- 9. Compare Linear Kernel, Polynomial Kernel, and RBF Kernel.**
The Linear Kernel is used when the data is linearly separable. The Polynomial Kernel works well for data with curved decision boundaries. The RBF Kernel is very flexible and is useful for complex non-linear data.
- 10. What is the effect of the C parameter in SVM?**
The C parameter controls the trade-off between a wide margin and correct classification. A large C gives less tolerance for misclassification, while a small C allows more errors but may improve generalization.
- 11. What is the role of the Gamma parameter in RBF Kernel SVM?**
Gamma controls how far the influence of a single training example reaches. A high gamma creates a very flexible boundary, while a low gamma gives a smoother boundary.
- 12. What is the Naïve Bayes classifier, and why is it called "Naïve"?**
Naïve Bayes is a probabilistic classifier based on Bayes' theorem. It is called "Naïve" because

it assumes that all features are independent of each other, which is usually not true in real life.

13. What is Bayes' Theorem?

Bayes' theorem describes how to update the probability of an event based on new evidence. In classification, it helps calculate the probability of a class given the input features.

14. Explain the differences between Gaussian Naïve Bayes, Multinomial Naïve Bayes, and Bernoulli Naïve Bayes.

Gaussian Naïve Bayes is used for continuous data and assumes a normal distribution. Multinomial Naïve Bayes is used for count-based features, such as word frequencies. Bernoulli Naïve Bayes is used for binary features, where values are 0 or 1.

15. When should you use Gaussian Naïve Bayes over other variants?

Gaussian Naïve Bayes should be used when the input features are continuous and approximately normally distributed.

16. What are the key assumptions made by Naïve Bayes?

Naïve Bayes assumes that all features are conditionally independent given the class label. It also assumes that the chosen probability distribution fits the data type.

17. What are the advantages and disadvantages of Naïve Bayes?

Its advantages are that it is fast, simple, and works well with high-dimensional data. Its disadvantages are that it makes a strong independence assumption and may perform poorly when features are highly correlated.

18. Why is Naïve Bayes a good choice for text classification?

Naïve Bayes works well for text because text data usually has many features, and the algorithm is efficient and effective for word frequency-based classification problems.

19. Compare SVM and Naïve Bayes for classification tasks.

SVM usually gives better accuracy on complex data and can handle non-linear boundaries using kernels. Naïve Bayes is faster and simpler, and it often performs well in text classification, but it is less powerful when feature independence does not hold.

20. How does Laplace Smoothing help in Naïve Bayes?

Laplace smoothing prevents zero probability problems by assigning a small probability to unseen feature values. This improves model stability and helps avoid failing predictions.